On minimal embedding of processes

(joint work with Mikhail Urusov)

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Outline

1. Introduction
2. Known results on Embeddings of processes
3. Embedding into a Brownian motion with initial integrable distribution

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1 Introduction

2 Known results on Embeddings of processes

3 Embedding into a Brownian motion with initial integrable distribution

Definitions

Let $B = (B_s)_{s \geq 0}$ be an adapted continuous (or càdlàg) stochastic process on a filtered probability space $(\Omega, \mathscr{F}, (\mathscr{F}_t)_{t \geq 0}, \mathrm{P})$.

Definition

1. *A finite $(\mathscr{F}_t)$-stopping time τ is said to be minimal for B if, for a stopping time σ, $\sigma \leq \tau$ and $B_\sigma \sim B_\tau$ imply $\sigma = \tau$ a.s.*
2. *A time change is a family $(T_s)_{s \geq 0}$ of $(\mathscr{F}_t)$-stopping times such that the maps $s \mapsto T_s$ are a.s. nondecreasing and right-continuous.*
3. *An embedding of a stochastic process X into a stochastic process B is a time change (T_s) such that the processes $(X_s)_{s \geq 0}$ and $(B_{T_s})_{s \geq 0}$ have the same law.*
4. *An embedding (T_s) of a stochastic process X into a stochastic process B is called minimal, if all stopping times T_s, $s \geq 0$, are minimal.*

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Stopping times with infinite values

If the process B has limit $B_\infty := \lim_{t\to\infty} B_t$ a.s., then one can consider embeddings with $[0,\infty]$-valued time change (T_s), i.e. with stopping times T_s, $s \geq 0$, with values in $[0,\infty]$.

Remarks on minimal stopping times

The concept of minimality is suggested in Monroe (1972) and attributed to Doob. It is proved in Monroe (1972) (in the case where B is a standard BM) that

- All stopping times with finite expectations are minimal.
- For any stopping time σ there is a minimal stopping time τ such that $\tau \leq \sigma$ and $B_\sigma \sim B_\tau$.
- Let τ be a stopping time such that $\mathrm{E}B_\tau = 0$. Then τ is minimal iff the family $(B_{t\wedge\tau})_{t\geq 0}$ is uniformly integrable.

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Classical SEP I

Problem (Embedding in a Brownian motion, naive formulation)

Given: a real-valued random variable ξ.
To find: a filtered probability space $(\Omega, \mathscr{F}, (\mathscr{F}_t)_{t\geq 0}, \mathsf{P})$, an $(\mathscr{F}_t, \mathsf{P})$-Brownian motion $B = (B_t)$ and a finite $(\mathscr{F}_t)$-stopping time τ such that $B_\tau \sim \xi$.

Problem I admits the following trivial solution. Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f(B_1) \sim \xi$. Then, with

$$\tau := \inf\{t \geq 1 : B_t = f(B_1)\},$$

due to recurrence of a Brownian motion, we have $B_\tau = f(B_1) \sim \xi$. This solution is attributed to Doob and is intended to show that without additional requirements the problem is trivial.

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SEP II

The original formulation of the SEP contains some restrictions:

Problem (SEP, Skorokhod (1961/1965))

Given: a real-valued random variable ξ *with* $\mathrm{E}\xi = 0$ *and* $\mathrm{E}\xi^2 < \infty$. *To find: a filtered probability space* $(\Omega, \mathscr{F}, (\mathscr{F}_t)_{t\geq 0}, \mathrm{P})$, *an* $(\mathscr{F}_t, \mathrm{P})$*-Brownian motion* $B = (B_t)$ *and an* $(\mathscr{F}_t)$*-stopping time* τ *with* $\mathrm{E}\tau < \infty$ *such that* $B_\tau \sim \xi$.

Note that the stopping time τ in the above solution is excluded because, for this stopping time, $\mathrm{E}\tau = \infty$ (unless f is the identity function, which is only possible when $\xi \sim N(0,1)$). Let us further note that $\mathrm{E}\tau < \infty$ implies $\mathrm{E}B_\tau = 0$ and $\mathrm{E}B_\tau^2 = \mathrm{E}\tau$, hence we need to assume $\mathrm{E}\xi = 0$ and $\mathrm{E}\xi^2 < \infty$ in the formulation when we have the requirement $\mathrm{E}\tau < \infty$.

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SEP III

These assumptions ($\mathrm{E}\xi = 0$ and $\mathrm{E}\xi^2 < \infty$) constitute the drawback of the formulation in Problem II. For example, the stopping times in the original Skorokhod's construction (Skorokhod (1961/1965)) require from ξ only to have a finite mean, but, as we have just seen, unless we assume a finite variance, it is no longer clear how to select "good" stopping times.
A very natural way to select "good" stopping times is to require them to be minimal (instead of requiring $\mathrm{E}\tau < \infty$).

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Let us note that Problem III is more general than Problem II in the sense that each solution of Problem II is a solution of Problem III (if $\sigma \leq \tau$ are solutions of Problem II, then $\mathsf{E}\tau = \mathsf{E}\sigma = \mathsf{E}\xi^2 < \infty$, i.e. $\sigma = \tau$ a.s., hence τ is minimal), but we do not assume $\mathsf{E}\xi = 0$ and $\mathsf{E}\xi^2 < \infty$ any longer.

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Embedding of processes: Dambis–Dubins–Schwartz Theorem

Theorem (Dambis (1965), Dubins & Schwartz (1965))

For every continuous local martingale $M = (M_s)_{s \geq 0}$ with $M_0 = 0$, there exists an embedding of M into a standard Brownian motion B.

The corresponding time change can be described explicitly.
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Embedding of processes: Monroe's Theorem (1972)

Theorem (Monroe (1972))

Let $M = (M_s)_{s\geq 0}$ be a martingale with $\mathsf{E}M_0 = 0$. Then there is a minimal embedding of M into a standard Brownian motion B.

Embedding of processes: Monroe's Theorem (1978)

Theorem (Monroe (1978))

A càdlàg process $X = (X_s)_{s\geq 0}$ is a semimartingale iff there is an embedding of X into a standard Brownian motion B.

Embedding into a geometric Brownian motion

Let

$$Z_t = e^{W_t - t/2}, \quad t \geq 0,$$

where $W = (W_t)_{t \geq 0}$ is a standard Brownian motion. The process Z is a geometric Brownian motion.

Theorem (Gushchin & Urusov (2015))

Let $X = (X_s)_{s \geq 0}$ be a càdlàg process. There is an embedding (resp. an embedding with $[0, +\infty]$-valued stopping times) of X into a geometric Brownian motion Z iff X is a strictly positive (resp. nonnegative) supermartingale.

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A natural question is: which processes admit a minimal embedding into a Brownian motion B?

Below we give an answer for integrable processes X. The reason why we deal only with integrable processes is that there is a convenient characterisation of minimal stopping times τ if $\mathrm{E}|B_\tau| < \infty$.

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Below we give an answer for integrable processes X. The reason why we deal only with integrable processes is that there is a convenient characterisation of minimal stopping times τ if $\mathsf{E}|B_\tau| < \infty$.

Criterion of minimality for non-centered distributions

Theorem (Cox & Hobson (2006))

Let $B = (B_t)_{t\geq 0}$ be a Brownian motion with $B_0 = 0$, and let T be a stopping time such that $B_T \sim \mu$, where μ is an integrable distribution with mean $m \geq 0$. Then the following conditions are equivalent:

- $\mathrm{E}|B_T| < \infty, \mathrm{E}B_T \geq 0$ *and* T *is minimal;*
- *the family* $\{B_S^+\}$ *is uniformly integrable, where* S *is an arbitrary stopping time such that* $S \leqslant T$
- $\mathrm{E}|B_T| < \infty$ *and* $\mathrm{E}B_{T\wedge H_c} \leqslant \mathrm{E}B_T$ *for any* $c > 0$, *where* $H_c = \inf\{t : B_t > c\}$

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Minimal embedding of integrable processes I

Theorem (Gushchin & Urusov (2019))

Let $X = (X_s)_{s \geq 0}$ be an integrable process, i.e. $\mathrm{E}|X_s| < \infty$ for each $s \geq 0$. Then there is a minimal embedding of X into a standard Brownian motion B iff

- *either X is a supermartingale with $\mathrm{E}X_0 \leq 0$*
- *or X is a submartingale with $\mathrm{E}X_0 \geq 0$.*

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The proof follows the same lines as the proof of the theorem in Monroe (1972) with intensive use of the criterion of minimality in Cox & Hobson (2006).

The previous theorem says that the class of integrable processes that can admit a minimal embedding into a Brownian motion starting from a deterministic point consists of all supermartingales and submartingales. This class can be enlarged if we embed into a Brownian motion starting from an some integrable distribution.

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Potential, integrated distribution and quantile functions

Recall that the potential of a probability measure μ is a function

$$u_\mu(x) = -\int |x-y|\mu(dy) = -\mathsf{E}|x-\xi|, \quad \xi \sim \mu$$

that is concave and 1-Lipschitz.

For a probability measure μ with distribution function $F_\mu(t) = \mu(-\infty; t]$, we define the integrated distribution function

$$J(x) := \int_0^x F_\mu(t)\,dt, \quad x \in \mathbb{R},$$

and the integrated quantile function is defined as the Fenchel transform of the integrated distribution function

$$K(u) := \sup_{x\in\mathbb{R}} \big\{xu - J(x)\big\}.$$

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The set $\mathcal{A}$ and auxiliary results

In order to state the criterion, we introduce the set

$$\mathcal{A} = \Big\{ x \in [-\infty, +\infty] : \lim_{y \to x} \big[u_\mu(y) - u_{\mu_0}(y) \big] = \sup_x \big[u_\mu(x) - u_{\mu_0}(x) \big] \Big\},$$

its boundary points $a_- = \inf \mathcal{A}, a_+ = \sup \mathcal{A}$ and hitting time $H_{\mathcal{A}} = \inf\{ t \geqslant 0 : B_t \in \mathcal{A} \}$

We now introduce two functions that will serve as the main tools for presenting the set $\mathcal{A}$ in another way:

$$H(x) := J_0(x) - J_1(x), \quad x \in \mathbb{R}$$

$$G(u) := K_1(u) - K_0(u), \quad u \in [0, 1].$$

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The set $\mathcal{A}$ and auxiliary results

Proposition

The following relations hold:

1. $\lim\limits_{x\to+\infty} H(x) = G(1), \qquad \lim\limits_{x\to-\infty} H(x) = G(0);$
2. $\sup\limits_{x\in\mathbb{R}} H(x) = \sup\limits_{u\in(0,1)} G(u).$

Proposition

Let $x^* \in \mathbb{R}$ be a point of global maximum of the function $H(x)$. Then the intersection of subdifferentials $\partial J_0(x^*) \cap \partial J_1(x^*)$ is non-empty and any level u^* from this intersection is a point of global maximum of the function $G(u)$, with $x^* \in \partial K_0(u^*) \cap \partial K_1(u^*)$.

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Proposition

The set $\mathcal{A}$ coincides with the set of points where the function H (including its limits at $\pm\infty$) attains its global supremum on the extended real line $[-\infty, \infty]$, that is,

$$\mathcal{A} = \left\{ x \in [-\infty, \infty] : H(x) = \sup_z H(z) \right\},$$

where for $x = \pm\infty$, $H(x)$ is understood as the corresponding limit $\lim_{y \to x} H(y)$.

The set $\mathcal{A}$ and auxiliary results

Denote the set $U^* := argmax_{u\in[0,1]} G(u)$

Theorem

A point $x \in [-\infty, \infty]$ belongs to the set $\mathcal{A}$ iff one of the following conditions holds:

1. *For finite points $x \in \mathbb{R}$: there exists a level $u \in U^* \cap (0,1)$ such that the subdifferentials of the integrated quantile functions intersect at the point x:*

$$x \in \partial K_0(u) \cap \partial K_1(u)$$

2. *For the point $x = +\infty$: the maximum of the function $G(u)$ is attained at the right boundary, that is, $1 \in U^*$.*
3. *For the point $x = -\infty$: the maximum of the function $G(u)$ is attained at the left boundary, that is, $0 \in U^*$.*

Criterion of minimal embedding into Brownian motion with integrable initial distribution

Theorem (Cox (2008))

Let $B = (B_t)_{t\geq 0}$ be a Brownian motion with initial integrable distribution $B_0 \sim \mu_0$, and let T be a stopping time such that $B_T \sim \mu$, where μ is an integrable distribution. Let the set $\mathcal{A}$ and the boundaries a_-, a_+ be defined as above Then the following conditions are equivalent:

- $\mathrm{E}|B_T| < \infty$ *and* T *is minimal;*
- $T \leqslant H_{\mathcal{A}}$ *a.s., and for any stopping times* R, S *such that* $R \leqslant S \leqslant T$,

$$\mathrm{E}(B_S \mid \mathscr{F}_R) \leqslant B_R \quad \text{a.s. on the set } \{B_0 \geqslant a_-\},$$
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Criterion of minimal embedding into Brownian motion with integrable initial distribution

Theorem (Cox (2008))

Let $B = (B_t)_{t\geq 0}$ be a Brownian motion with initial integrable distribution $B_0 \sim \mu_0$, and let T be a stopping time such that $B_T \sim \mu$, where μ is an integrable distribution. Let the set $\mathcal{A}$ and the boundaries a_-, a_+ be defined as above Then the following conditions are equivalent:

- $\mathrm{E}|B_T| < \infty$ *and* T *is minimal;*
- $T \leqslant H_{\mathcal{A}}$ *a.s., and for any stopping times* R, S *such that* $R \leqslant S \leqslant T$,

$$\mathrm{E}(B_S \mid \mathscr{F}_R) \leqslant B_R \quad \text{a.s. on the set } \{B_0 \geqslant a_-\},$$
$$\mathrm{E}(B_S \mid \mathscr{F}_R) \geqslant B_R \quad \text{a.s. on the set } \{B_0 \leqslant a_+\}.$$

Minimal embedding of integrable processes II

Theorem

Let $X = (X_s)_{s\geq 0}$ be an integrable process, i.e. $\mathsf{E}|X_s| < \infty$ for each $s \geq 0$. Then there is a minimal embedding of X into a Brownian motion $B = (B_s)_{s\geq 0}$ with $B_0 \sim \mu_0$ on $\mathbb{R}$ iff there are a_-, $a_+ \in [-\infty, +\infty]$, $a_- \leq a_+$, such that

- *conditionally on the set $\{X_0 \leq a_-\}$, X is a submartingale with values in $(-\infty, a_-]$;*
- *conditionally on the set $\{X_0 \geq a_+\}$, X is a supermartingale with values in $[a_+, +\infty)$;*
- *conditionally on the set $\{a_- < X_0 < a_+\}$, X is a martingale with values in $[a_-, a_+]$.*

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Thank you!